



AFRL

Modern Unmanned-warfare Systems Environment (MUSE)

Industry Day

AFRL/RIGD, 17 JUN 2025

**Brendon Poland, AFRL/RI, C-UAS Technical Director
Matthew Zawisza, AFRL/RI, C-UAS Acquisition Program Manager
Meghan Thomas, AFRL/RI, Contracting Officer**

Agenda

- 0800-0900 Sign In
- 0900-0910 Introduction – Matt Zawisza
- 0910-0920 Acquisition Background – Meghan Thomas
- 0920-0930 AFRL C-UAS Overview – Phil Zaleski
- 0930-1030 C-UAS Technology Areas – Brendon Poland
- 1030-1100 Q&A

Lunch Break

- 1200 Break-out Sessions/20-minute blocks
- 1200-1600 Industry Networking (Event Space)

Purpose

- General Session: Provide interested parties an overview of AFRL's role in C-UAS and the scope of the Modern Unmanned-warfare Systems Environment (MUSE) Program
 - Overview of where we are going
 - Overview of current systems
 - Technical deep dive
 - Open forum for Q&A
 - Opportunity to meet with potential teaming partners
- Afternoon: One-on-One Sessions
 - Scheduled for 20 minutes max
 - **AFRL is not requesting, nor will accept any marketing material**

Ground Rules

- No audio or video recording is permitted
- AFRL will publish responses to all relevant questions from the presentations and the one-on-ones and will provide answers through SAM.gov in the near future
- If you have any additional questions, please submit them NLT 1200 pm EST on 24 JUN 2025, and
- To be added to the Interested Parties List posted on SAM, please use the following email addresses:

matthew.zawisza@us.af.mil

meghan.thomas.1@us.af.mil

patrick.cerminaro@us.af.mil

MUSE Technical Read Library

- Available now
- The library contains technical data subject to export control
- Potential Offerors who wish to gain access will be required to
 - furnish a copy of their company's DD Form 2345 - Militarily Critical Technical Data Agreement
 - provide a letter from their Data Custodian designated on that form authorizing access by whomever will be reviewing the library
 - sign a Non-Disclosure Agreement (NDA)
- If interested in access to the technical library, please contact Matthew Zawisza by email at matthew.zawisza@us.af.mil and AFRL.CUAS.PMO@us.af.mil

MUSE Acquisition Overview

Meghan Thomas

Acquisition Team

- AFRL/RI C-UAS Branch Chief – Phil Zaleski
- AFRL/RI C-UAS Technical Director – Brendon Poland
- AFRL/RI C-UAS Lead Program Manager – Kyle Holbritter
- AFRL/RI C-UAS Acquisition Program Manager – Matthew Zawisza
- AFRL/RI Contracting Officer – Meghan Thomas
- AFRL/RI Contract Specialist – Patrick Cerminaro

Acquisition Overview

- Maximum Ordering Amount: ~\$500M
- Estimated IDIQ Labor Scope:
 - Approximately 3.4M hours
- Anticipated Period of Performance:
 - Anticipated Ordering Period: 60 Months
- Principal Purpose: R&D, to include O&M, R&D, and Procurement Requirements
- Type of Contract: Single award Indefinite Delivery Indefinite Quantity (IDIQ) with Cost Plus Fixed Fee (CPFF) (Level of Effort (LOE) and Completion Basis) and Firm Fixed Price (FFP) Task Orders
 - Order type to be determined on an order-by-order basis, based on the nature of the requirements

Market Research

- Request for Information (RFI): Posted to SAM.gov between the dates of 19 May 2025 and 03 June 2025
- Intend to release Sources Sought in 4th quarter of CY 2025
 - The determination for a small business set-aside will be established based on the submissions to the Sources Sought
- Intend to release DRAFT RFP during 1st quarter of CY 2026, via SAM.gov
 - DRAFT RFP will include anticipated first order
 - Industry will have opportunity to comment/provide feedback in response to DRAFT RFP
- Contact with myriad of potentially-interested companies, via combination of in-person meetings, phone conversations, email correspondence, and accessing of the Technical Read Library

Source Selection Procedures

- Anticipate use of Subjective Trade Off methodology, as contemplated in DOD Source Selection Procedures, for evaluation of proposals
- Government values superior technical features/capabilities greater than lowest cost/price to the Government
- May award with or without discussions, as determined appropriate considering circumstances
- Prime to be selected based on at least one (1) real order, to be included in solicitation

Acquisition Milestones

<u>Milestone</u>	<u>Date</u>
• Pre-Solicitation Notice	May 2025
• Tech Read Library	Jun 2025
• Industry Day	Jun 2025
• Sources Sought	Oct 2025*
• Draft RFP Release	Jan 2026*
• RFP Release	July 2026*
• Proposals Due	Aug 2026*
• Contract Award	Aug 2027*
• Note: Basic IDIQ to be awarded in conjunction with an initial order	

*These dates are early estimates and are subject to change as the acquisition strategy is developed

Program History

- FA8750-16-C-0206 PAR Tech: Aerial Layer to a Cyber City (ALCC)
- FA8650-16-C-9205 TSC: Support Jammer Cueing (SJC) Rapid Reaction Technologies & Systems (RRTAS)
- FA8750-16-C-0293 AX Enterprize: Collaborative Low-Altitude UAS Integration Effort (CLUE)
- FA8650-17-C-9205 (SBIR II) Black River Systems: CUAS Open Systems Architecture (COSA)
- FA8650-17-C-9206 (SBIR II) DZYNE: Counter Unmanned Rapid Engagement Systems (CURES)
- FA8650-17-C-9207 (SBIR II) Sarcos: Small Drone Defeat System (S2DS)
- FA8650-17-C-9208 (SBIR II) Fortem: Mitigation of Small Unmanned Aircraft Systems
- FA8750-17-S-7001 (BAA Topic): Cyber Asymmetric Force Applications for UAS
- FA8750-17-C-0196 Raytheon: Vulnerability Analysis & Exploitation
- FA8750-17-C-0190 Steven Institute Contract: Acoustic Methods for UAS Detection, Tracking and Localization
- FA8750-17-C-0197 Black River Systems: Unmanned Aircraft Systems Asymmetric Cyber Applications (UASACA)
- FA8750-18-C-0041 KRI/Northeastern University: Unmanned Aerial Systems (UAS) Detection and Counter UAS R&D
- FA8750-19-C-0040 (SBIR III) Black River Systems: Operational Counter-sUAS Open Systems Architecture (OCOSA)
- FA8750-21-D-1100 (IDIQ) PAR Government Systems: Counter-small Unmanned Aircraft Systems (C-sUAS)
- FA8650-23-C-9205 (SBIR II) Trust Automation: Spectrally Responsible Counter-PNT System
- FA8650-23-C-9206 (SBIR II) Black River Systems: MP-Jammer for Responsible Navigation Warfare
- FA2385-24-C-B028 (SBIR II) Zone 5 Technologies: Counter-UAS Long Bow
- FA2385-24-C-B029 (SBIR II) Decryptor: Counter-UAS Long Bow

AFRL C-UAS Overview

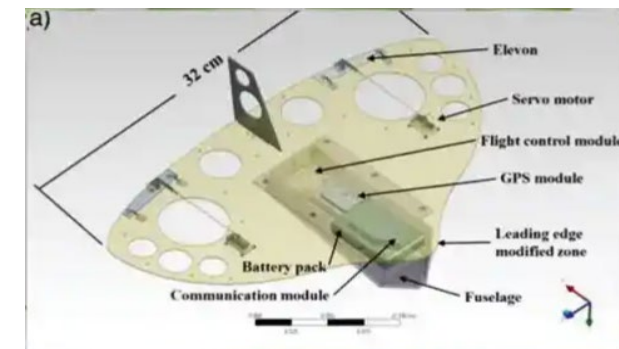
Phil Zaleski

The UAS Threat (On the move again)

- ISIS (2016-2019)
 - [What We Know About ISIS's Scratch-built Drones | Popular Science \(popsci.com\)](https://www.popsci.com/isis-drones/)
 - [Drones are biggest tactical concern since the rise of IEDs in Iraq, CENTCOM boss says \(militarytimes.com\)](https://www.militarytimes.com/news/your-story/2016/08/01/drones-are-biggest-tactical-concern-since-the-rise-of-ieds-in-iraq-centcom-boss-says/)
- Future of Global Terrorism (2021)
 - <https://www.ausa.org/publications/role-drones-future-terrorist-attacks>
 - [International Expert Group Meeting on Vulnerable Targets and Unmanned Aircraft Systems | Office of Counter-Terrorism](https://www.ofcrt.org/International-Expert-Group-Meeting-on-Vulnerable-Targets-and-Unmanned-Aircraft-Systems)
- Ukraine (2022)
 - <https://www.securityinfowatch.com/perimeter-security/robotics/anti-drone-technologies/article/21277392/weaponization-of-commercial-drones-is-a-global-threat>
 - [Small Drones Are Giving Ukraine an Unprecedented Edge | WIRED](https://www.wired.com/story/small-drones-are-giving-ukraine-an-unprecedented-edge/)
- China (2023-Future)
 - [China Sees Balloon-Launched Drone Swarms In Its Future | The Drive](https://www.theverge.com/2023/3/28/23611111/china-sees-balloon-launched-drone-swarms-in-its-future)
 - [China's Autonomous Attack Drones are Ready to Take Off | The National Interest](https://www.nationalinterest.org/technology/chinas-autonomous-attack-drones-are-ready-to-take-off-144881.php)

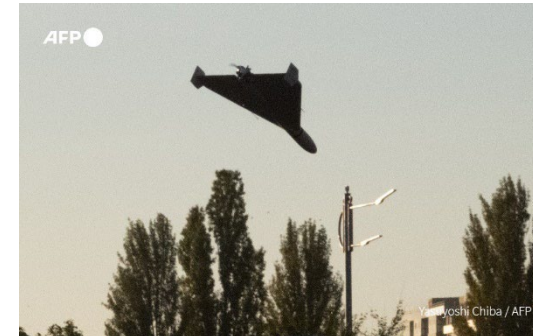


A small ISIS drone, captured by Iraqi police, rests on a table at an intelligence-sharing meeting at Qayyarah West Airfield, in Iraq, (Jason Hull/Army)



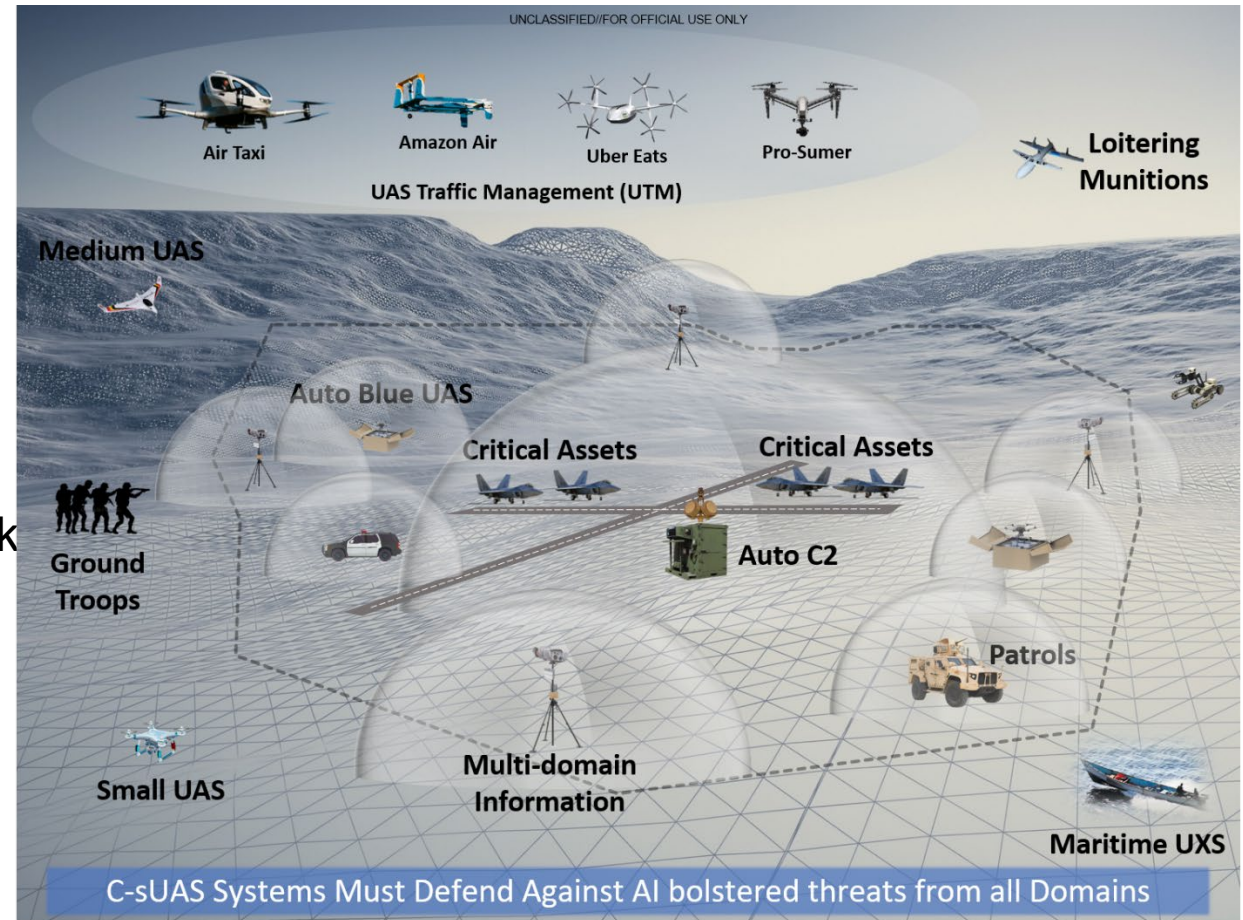
Technical Trends and Challenges

- Threats are growing in speed and capability to carry additional explosives but still evade sensors
- Threats are trending toward more autonomy to avoid communication links and jamming
- Threats are faster and fly slower to avoid early detection and track
- Counter systems must be able to detect, ID and track low and slow with near zero false alarm
- Systems must run the full kill chain at computer speed and not violate rules of engagement
- Systems must be effective against one and many attacking UAS



Areas of S&T opportunity

- Precision Electronic Attack
 - C2 Jamming doesn't stop real threats
 - Attacking Control vs Communications
- Counter Position Navigation & Timing
 - Effective but heavy handed
 - Responsible PNT Jamming
- Interceptors
 - Kinetics in permissive environments work
 - Non-permissive and automated UAS
- System of Systems
 - Full automation
 - Trusted data sources to automate
 - Consideration of "legal traffic"



AFRL C-UAS Technology Areas

Brendon Poland

Technical Objectives

1. Equip the US government with technical tools to combat the full spectrum of threats
2. Partner with industry to bring this technology to bear at wartime speed
3. Demand near symmetry to our costs and timelines in comparison to the threat
4. Balance “Cost Speed Quality” paradigm on a task-by-task basis

Technology development strategy

- Agile development of C-UAS prototypes and capabilities that fill technology gaps across the entire C-UAS kill-chain
 - Leverage existing and enabling technology to focus teams on non-realized capabilities
 - Layers of complementary and efficiently paired technology
 - Quickly assemble new teams with new ideas to tackle hard problems
 - Complementary Government and contractor engineering teams
- Early experimentation and validation in near operational environments
 - Our DoD facilities are our test sites
 - Our Security Forces, Police and Defenders are our customers
- An acquisition that can move technology from basic research all the way to fielded systems due to proof of the technology's utility in operational environments

Technical Focus Areas

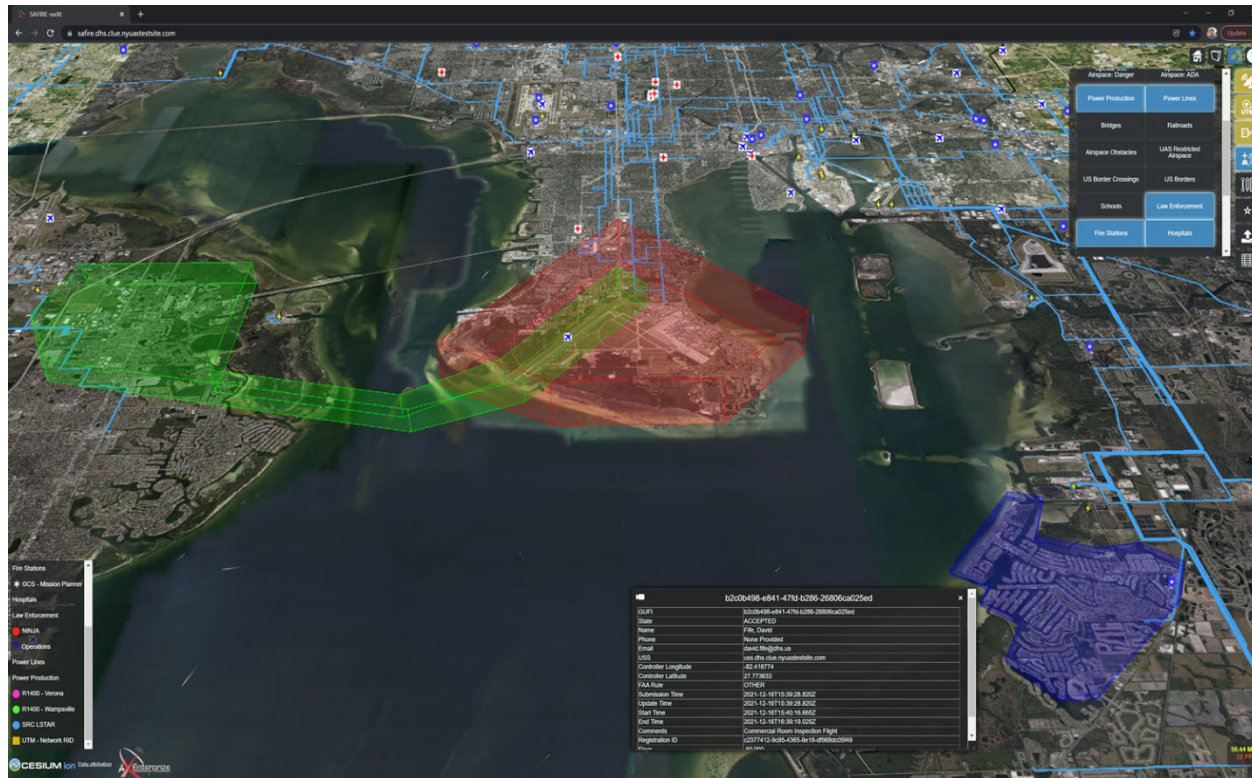
- EW Enabled cyberspace countermeasures (CM)
- Low collateral damage (CD) Electronic Attack/CM
- C-sUAS operations in RF/Cyberspace contested environment
- Defensive Cyber Space Operation (DCO)
- Offensive Cyberspace Operations (OCO)
- Cyberspace C2
- Cyberspace Intelligence, Surveillance, and Reconnaissance (ISR)
- Automation and Autonomy of C2 systems
- Multi-domain C2 (MDC2)
- Airspace Coordination Measures (ACM)/Air Traffic Control (ATC)
- Unmanned Traffic Management (UTM) Systems
- Blue Force Interceptors
- Airborne C-sUAS C4ISR
- Directed Energy Weapon development and integration
- Sensor development and integration
- Tactical and Strategic tipping and cueing
- Signature and Effects Library

DIVERSE PROGRAM REQUIRING A BROAD SET OF SKILLS

UAS Traffic Management

- Enable users to utilize UTM capabilities, while addressing needs and requirements specific to them (i.e., security, information sharing, etc.)

SAFIRE-X C2



DART C2 for Threat Assessment



Ninja Family of Systems



NinjaGen2



NinjaTRX



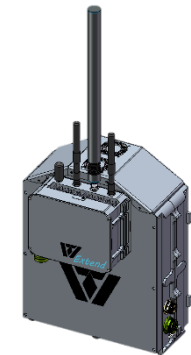
NinjaES



**Ninja Dismount
(Prototype)**



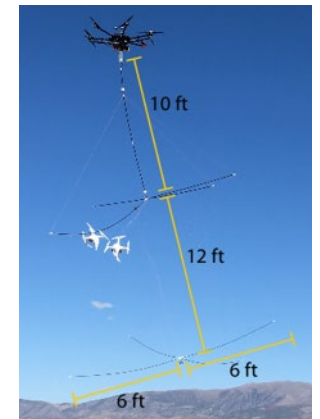
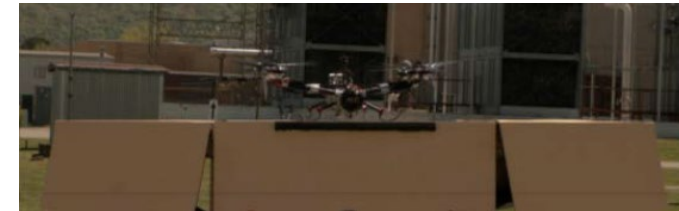
Windtalker V2



**Windtalker V3
XTEND**

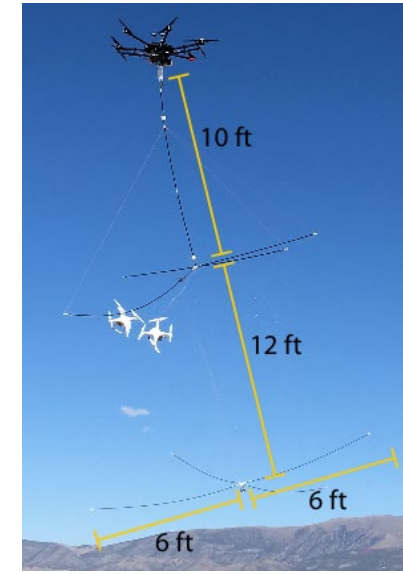
Low Collateral Effects Interceptor

- Engineering Analysis of each leading interceptor
- Mature systems in preparation for acquisition decisions by the services
- Deliver working systems, performance data and documentation
- Systems need to autonomously engage many targets with minimal operator intervention



Paladin Multi-Mission UAS

- Fully autonomous “wingman” for security operations
- Full installation security suite of missions (i.e. perimeter security, dynamic target tracking, UAS interdiction)
- Build the system on the best and most affordable commercial UAS
 - Reduces cost
 - The science is the algorithms, AI and weaponneering
- System requires cue from other drone tracking system
 - Currently integrated with DoD Radars, Ninja etc
- Onboard Radar/Camera and AI guides vehicle to intercept
- SA by ground personnel through encrypted comms and ATAK



Tracking
Systems



Silvercrown Group 3 effort

- Full sharing and re-use of Group 2 interceptor sensing, software and hardware
- Leverage industry investment in UAS
- Build a vertical take-off forward flight interceptor that can bear modular payloads

- Target Specifications

Cruise Speed: 85 knots

Sprint Speed: 110 knots

GVW: 80 lbs.

Payload: 25 lbs.

Wing Span: 80 inches

Weapons: Modular (Projectile, direct strike drones)

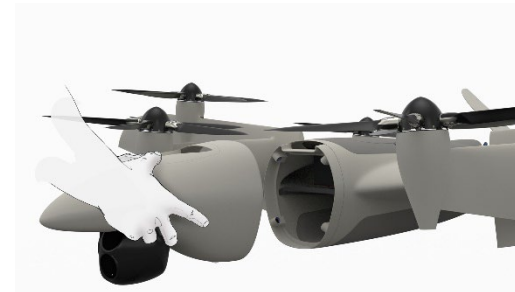
Sensors: Modular (Camera, Radar, Lidar)



Current Prototype



Modular Payloads

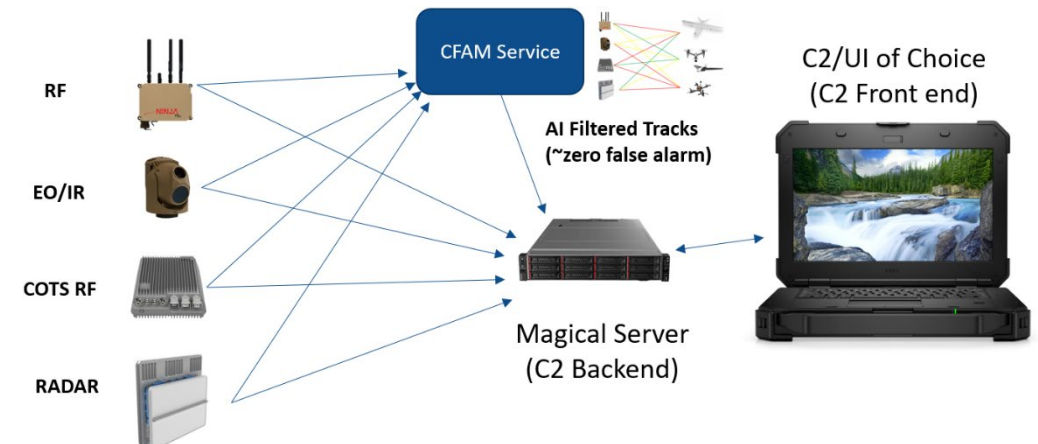
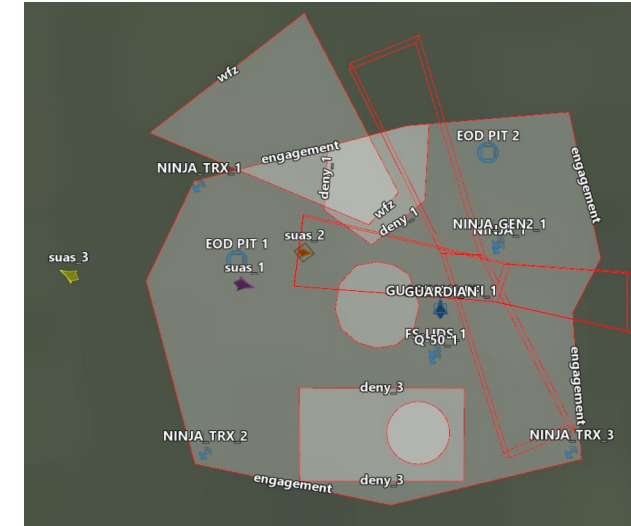


Tactical Employment



C2 Autonomy (C2A) & C-UAS False Alarm Mitigation (CFAM)

- Services needed for a fully functional System of System
- C2A Agent
 - Identifies the Rules of engagement
 - Identifies areas that have different priorities and rules
 - Decision tree pruning to develop courses of actions that are optimal and legal in seconds
 - Enables “One Button” C2
- CFAM Agent
 - Using many sensors to gain trust in UAS tracks
 - Enables automated C2 systems to validate their data
 - Systems get better with time
 - Requires human supervision of system not operation
 - Helps build the DoD repository of validated drone data



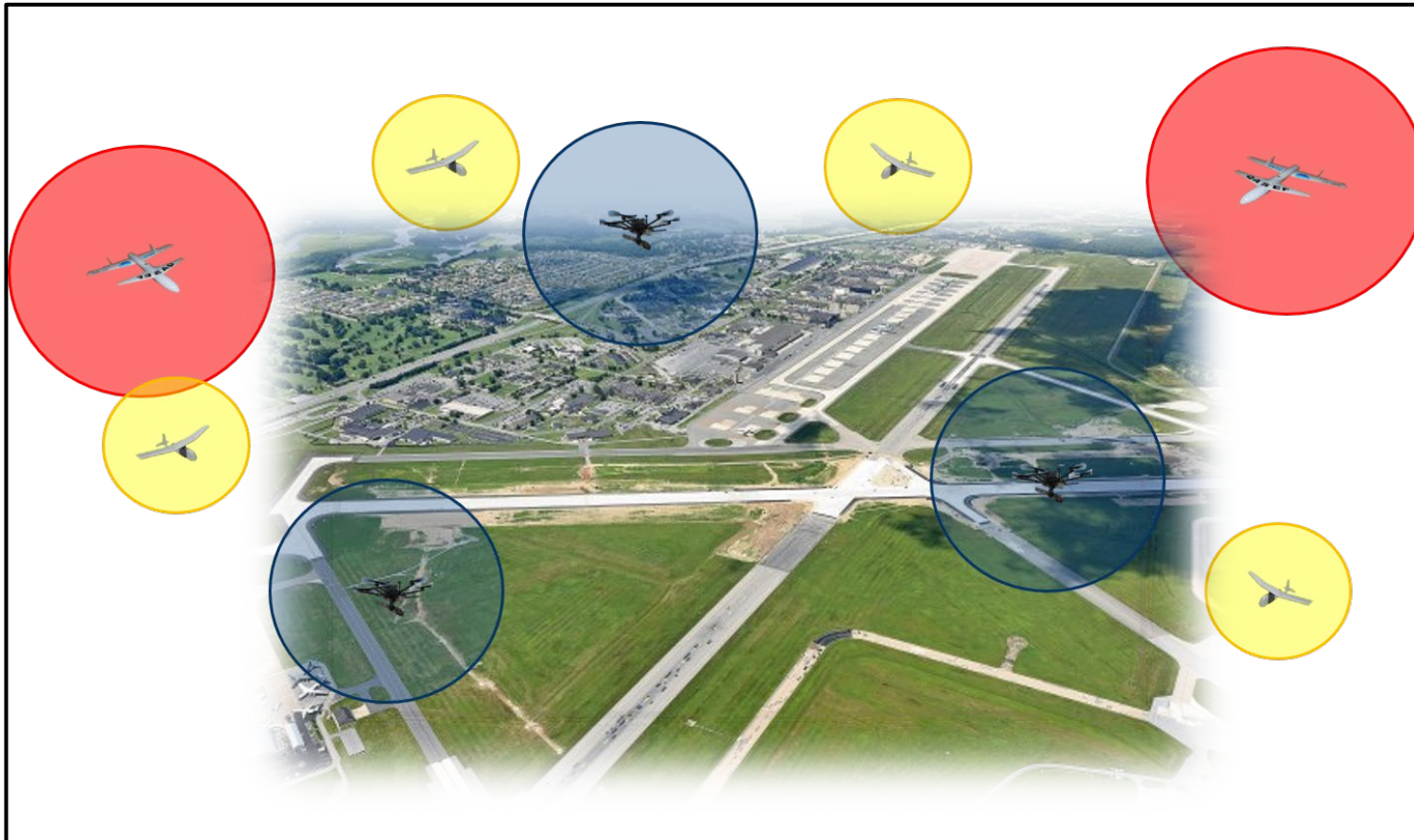
ACE Pallet System

- Full technical demonstrator of all needed technologies to detect and defeat drones
- System is designed to be rapidly deployable for protection of short-term operations
- Container space provides storage while in motion and operations center space when positioned for use
- System keeps users aware of airspace while away from the container via ATAK and Mesh radios



Mitigation of Unmanned Aircraft Systems

- Builds on all past EW/Cyber/C2 and AI/ML work to be ready for the emerging threat
- “UAS Battle School” concept



Summary

- **Technical Objectives**

1. Equip the US government with technical tools to combat the full spectrum of threats
2. Partner with industry to bring this technology to bear at wartime speed
3. Demand near symmetry to our costs and timelines in comparison to the threat
4. Balance “Cost Speed Quality” paradigm on a task-by-task basis

- **Technical Development Strategy**

1. Agile Development
2. Early Experimentation and Operational Evaluations
3. End to End RDT&E (No Valley of Death)

- **Tech Areas**

1. EW to Cyber
2. Autonomous Command and Control
3. Sensor Exploitation and integration
4. Interceptors to Directed Energy



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QUESTIONS?